

Problem-Solution Fit

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The Problem

Manual loan eligibility checks are slow, inconsistent, and offer no instant feedback to applicants.

The Solution

An ML-powered web application that takes applicant details as input and instantly predicts loan eligibility using a model trained on historical loan approval data.

Fit Analysis

Problem Aspect	How Smart Lender Addresses It
Slow manual review	Instant prediction (seconds vs days)
Inconsistent decisions	Same model logic applied uniformly to every applicant
No early feedback for applicants	Self-service web form gives an immediate estimate
Officer workload	Pre-filters straightforward cases, freeing time for edge cases

Validation Approach

- Model accuracy validated against a held-out test set (20% split).
- Compared multiple algorithms (Decision Tree, Random Forest, KNN, XGBoost) to pick the best performer.
- Manual sanity-checks using realistic sample applicant profiles.

Risks / Limitations

- Model is only as fair as the historical data it is trained on.
- Not a replacement for full regulatory underwriting in a real banking context.